

CDAC Architecture: Binary-Segmented with 2C Bridge

Anchor Wall
 $C9+CR=512$ LSB
(uncalibrated)

Calibration Targets
 $C10(512), C11(1024), C12(2048)$

Comparator
(1 LSB terminal)

High Segment
(64 Cu total)

$H=65$ LSB, weights:

65, 65, 130, 260, 520, 1040, 2080

1C-A 1C-R 2C 4C 8C 16C 32C

Bottom-Plate Switches $\rightarrow$ BITD[13:7] / BITU[13:7]

Bridge
Low Segment
(63 Cu total)

Weights: 2, 4, 8, 16, 32, 64 LSB

1C 2C 4C 8C 16C 32C

Bottom-Plate Switches $\rightarrow$ BITD[6:1] / BITU[6:1]

calDAC ($\Sigma=126$ LSB)
used as SAR search mechanism